

Measurement of Absolute Single Differential Cross-Section in p_T Bins using Runs 2 and 3 Data

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- Events from runs 2 and 3
- Kinematic Phase Space
- Event Selection Criteria
- Yield Distributions
- Efficiency Corrections
- Corrected Yields
- Corrected Yields Background Subtracted
- Single Differential Cross-Section Plots
- Next Steps

Events from runs 2 and 3 (Input Files Used)

Note: Currently using all the runs saved in runs 2 and 3.

Harsha's ROOT files:

/sequest/users/harshaka/e906.project/e906-root-ana/work_gpvm/scripts/results_runFinalTree/merged_results_e906_mixing/:

- merged_RS57_LH2_1_1138.root
- merged_RS57_LD2_3_1138.root
- merged_RS57_Empty_2_1138.root
- merged_RS59_LH2_1_465.root
- merged_RS59_LD2_3_466.root
- merged_RS59_Empty_2_466.root
- merged_RS62_LH2_1_1234.root
- merged_RS62_LD2_3_1237.root
- merged_RS62_Empty_2_1234.root
- merged_RS70_LH2_1_264.root
- merged_RS70_LD2_3_266.root
- merged_RS70_Empty_2_267.root

Abi's ROOT files:

/sequest/users/apun/e906_projects/rs67_merged_files/:

- merged_RS67_3089LH2.root
- merged_RS67_3089LD2.root
- merged_RS67_3089flask.root

Kinematic Binning Definition:

- **Mass Bins (GeV):** [4.2, 4.5, 4.8, 5.1, 5.4, 5.7, 6.0, 6.3, 6.6, 6.9, 7.5, 8.7]
- **p_T Bins (GeV):** [0.0, 0.32, 0.49, 0.63, 0.77, 0.95, 1.18, 1.8]
- **x_F Bins:** [0.0, 0.85] with a fixed step width of 0.05.

Event Selection Criteria (Chuck Cuts)

Dimuon Kinematics:

- $4.2 < \text{Mass} < 8.8 \text{ GeV}$, $-0.1 < x_F < 0.95$, $0.05 < x_T \leq 0.58$, and $|\cos \theta| < 0.5$.
- Vertex cuts: $-280 < dz < -5$, target/dump tracking separation criteria met.
- Transverse momentum bounds: $|dp_x| < 1.8$, $|dp_y| < 2.0$, and $dp_x^2 + dp_y^2 < 5$.

Track Quality & Acceptance:

- $\chi^2_{\text{target}} < 15$, $\chi^2_{\text{dimuon}} < 18$, and $\chi^2/(\text{nHits} - 5) < 12$.
- Hit requirements: $\text{nHits} > 13$ per track, $\text{nHits}_1 + \text{nHits}_2 > 29$, $\text{St1 total} > 8$.
- Track vertex longitudinal limits: $-320 < z_v < -5$.
- A dynamic beam offset correction is applied to y-coordinates: offset is 1.6 for $\text{runID} \geq 11000$ and 0.4 otherwise.

Detector Occupancy Limits:

- Drift chamber hits: $D1 < 400$, $D2 < 400$, $D3 < 400$, and total $D1 + D2 + D3 < 1000$.
- $20 < D1 < 385$ (global reconstruction efficiency curve defined in this range)

For RS62, $\text{runID} > 11500$ was applied to select dimuons with the correct target position Note: Currently using the same set of Chuck Cuts defined in DocDB 2111-V42

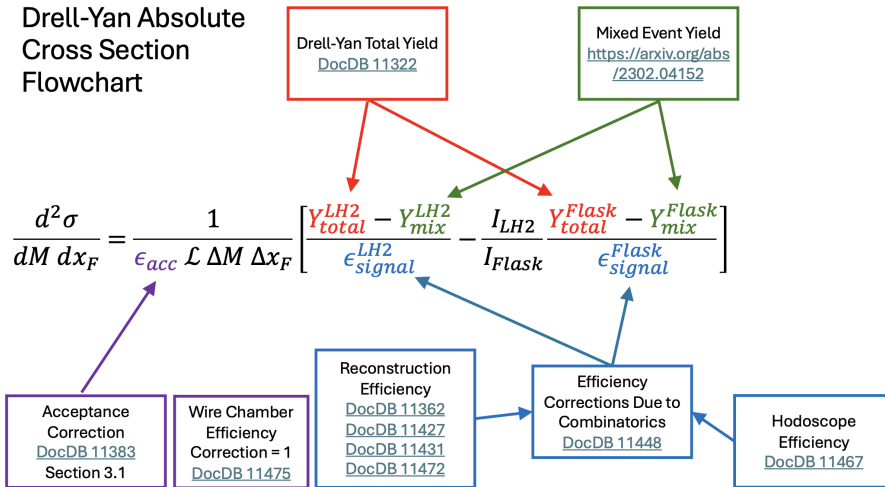
POT Values Used

Roadset	LH2	LD2	Flask
57	3.533324×10^{16}	1.768358×10^{16}	3.918550×10^{15}
59	9.365986×10^{15}	4.319952×10^{15}	1.010350×10^{15}
62	5.281659×10^{16}	2.382899×10^{16}	1.097456×10^{16}
67	1.611435×10^{17}	7.694541×10^{16}	3.662417×10^{16}
70	1.785745×10^{16}	8.752588×10^{15}	3.841280×10^{15}

Note: POT Values from runs mixed by Harsha in DocDB 11524.

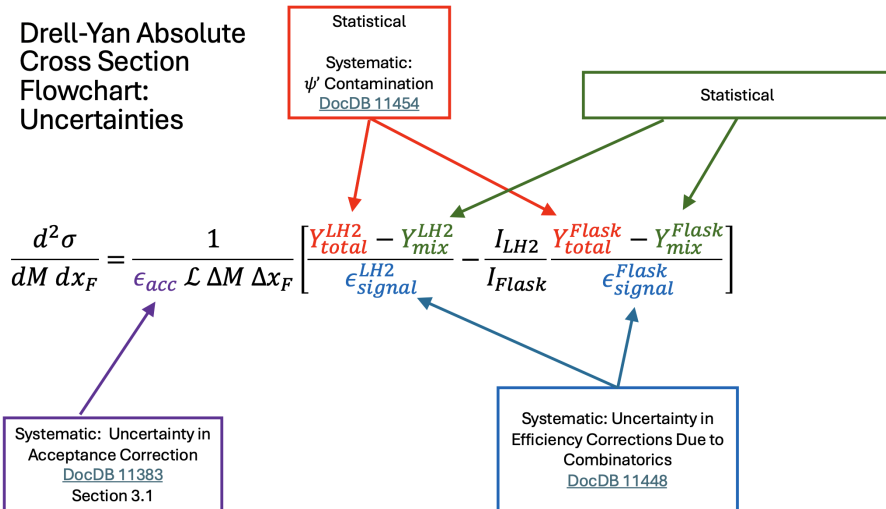
Analysis Procedure

Drell-Yan Absolute Cross Section Flowchart

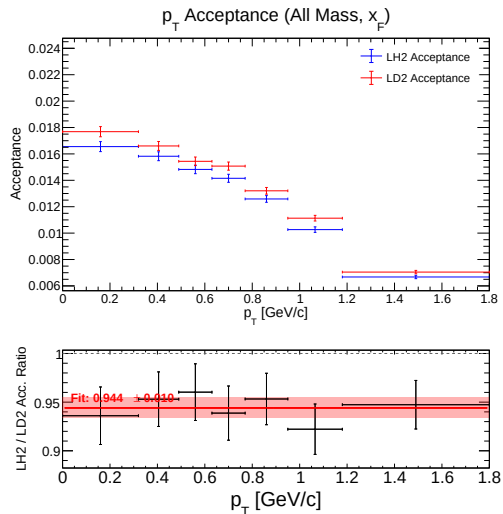


Uncertainties

Drell-Yan Absolute Cross Section Flowchart: Uncertainties



Acceptance Corrections for p_T bins



Determination of ϵ_{signal}

For a given target (LH2 or LD2 or Flask) and a given kinematic (M, x_F, p_T) bin, there will be Y_{total} dimuons in the normal event stream and Y_{mix} dimuons from the mixed event stream.

$$\langle \epsilon \rangle_{\text{total}}^{\text{hodo}} = \frac{1}{Y_{\text{total}}} \sum_{i=1}^{Y_{\text{total}}} \epsilon_i^{\text{hodo}}$$

$$\langle \epsilon \rangle_{\text{mix}}^{\text{hodo}} = \frac{1}{Y_{\text{mix}}} \sum_{j=1}^{Y_{\text{mix}}} \epsilon_j^{\text{hodo}}$$

$$\langle \epsilon \rangle_{\text{total}}^{\text{reco}} = \frac{1}{Y_{\text{total}}} \sum_{i=1}^{Y_{\text{total}}} \epsilon_i^{\text{reco}}$$

$$\langle \epsilon \rangle_{\text{mix}}^{\text{reco}} = \frac{1}{Y_{\text{mix}}} \sum_{j=1}^{Y_{\text{mix}}} \epsilon_j^{\text{reco}}$$

$$\epsilon_{\text{signal}} = \frac{\langle \epsilon \rangle_{\text{total}} Y_{\text{total}} - \langle \epsilon \rangle_{\text{mix}} Y_{\text{mix}}}{Y_{\text{total}} - Y_{\text{mix}}}$$

DocDB 11448

$$\epsilon_{\text{hodo}} = [\epsilon_{st1}^x \epsilon_{st2}^x \epsilon_{st3}^x \epsilon_{st4}^x \epsilon_{st1}^y \epsilon_{st2}^y \epsilon_{st3}^y \epsilon_{st4}^y]_i$$

$\delta \langle \epsilon \rangle_{\text{total}}^{\text{hodo}}$ and $\delta \langle \epsilon \rangle_{\text{mix}}^{\text{hodo}}$ are propagated from hodoscope paddle efficiencies.

ϵ^{reco} interpolated from $\epsilon(D1)$ function.

$$\langle \epsilon \rangle_{\text{total}} = \langle \epsilon \rangle_{\text{total}}^{\text{hodo}} \langle \epsilon \rangle_{\text{total}}^{\text{reco}}$$

$$\langle \epsilon \rangle_{\text{mix}} = \langle \epsilon \rangle_{\text{mix}}^{\text{hodo}} \langle \epsilon \rangle_{\text{mix}}^{\text{reco}}$$

$\delta \langle \epsilon \rangle_{\text{total}}^{\text{reco}}$ and $\delta \langle \epsilon \rangle_{\text{mix}}^{\text{reco}}$ include correlations

(See DocDB 11431)

$\delta \epsilon_{\text{signal}}$ propagated from uncertainties in the six inputs.

Following Yields were calculated for targets: [LH2, LD2 and Flask]:

LH2 Target:

- `Y_total_LH2, Y_mix_LH2`

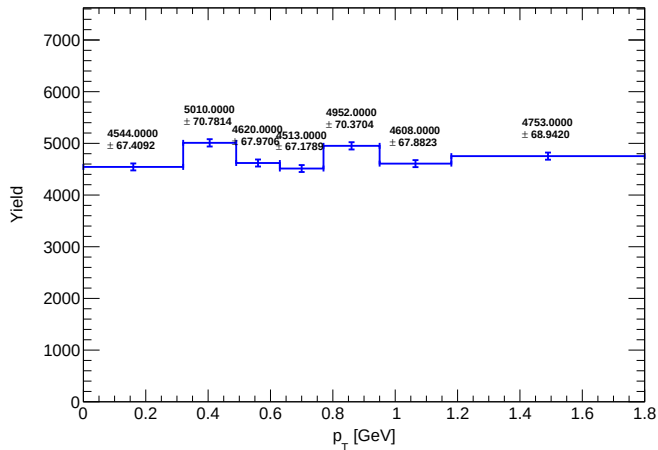
LD2 Target:

- `Y_total_LD2, Y_mix_LD2`

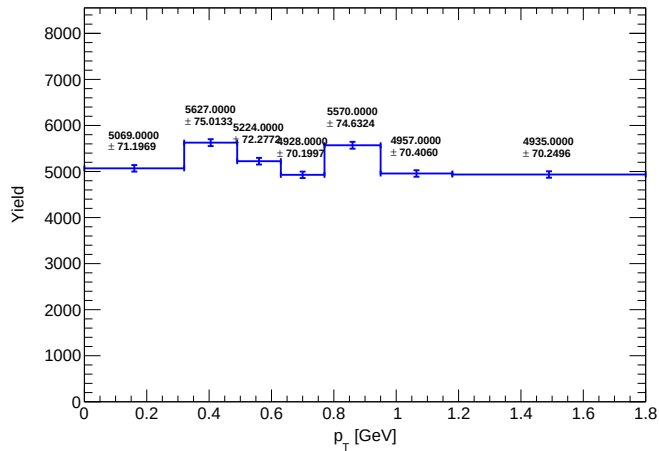
Flask:

- `Y_total_Flask, Y_mix_Flask`

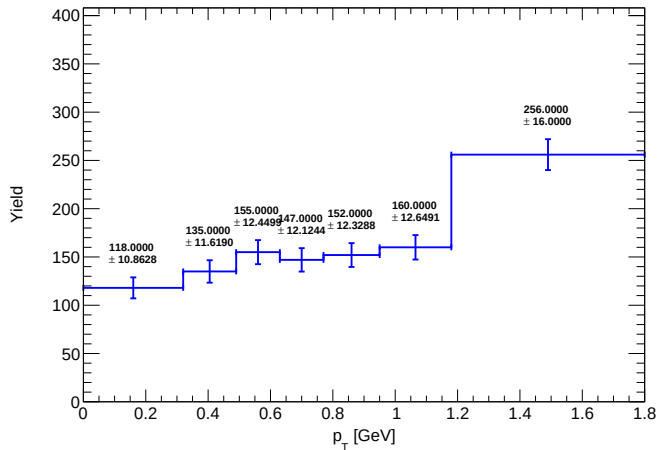
Total Yield (result) (LH2)



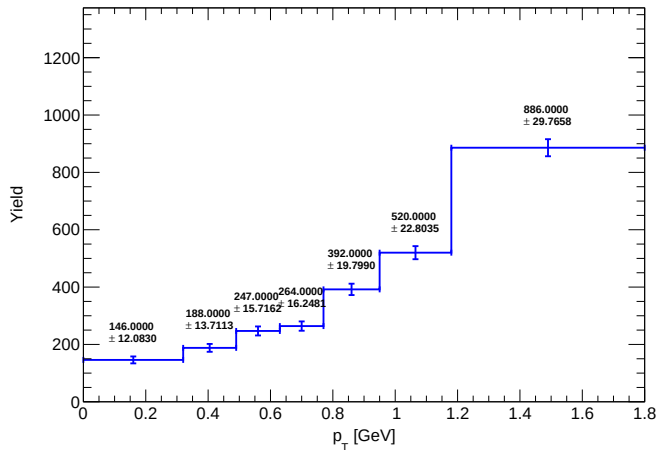
Total Yield (result) (LD2)



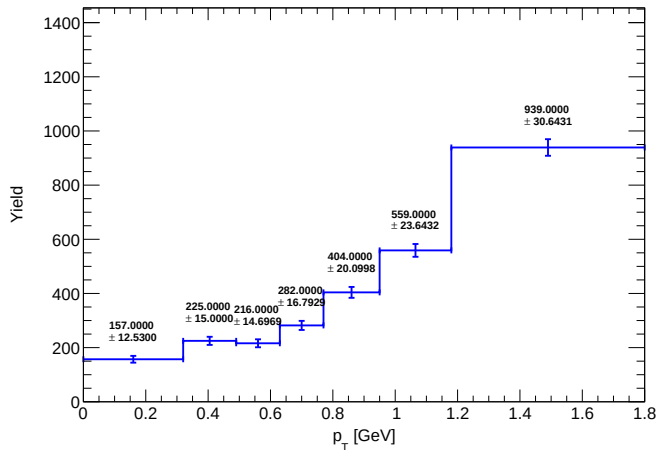
Total Yield (result) (Flask)



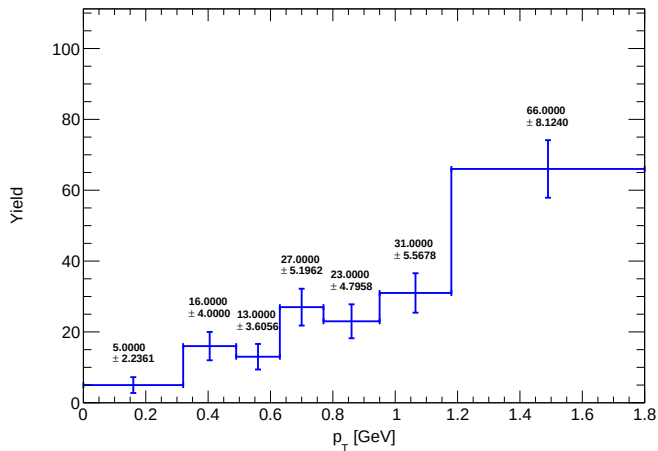
Mix Yield (result_mix) (LH2)



Mix Yield (result_mix) (LD2)



Mix Yield (result_mix) (Flask)



Determination of reconstruction efficiency

Following reconstruction efficiencies with the propagated uncertainties were calculated for targets: [LH2, LD2 and Flask] by using D1 occupancy and the global efficiency curve:

LH2 Target:

E total reco LH2, E mix reco LH2

LD2 Target:

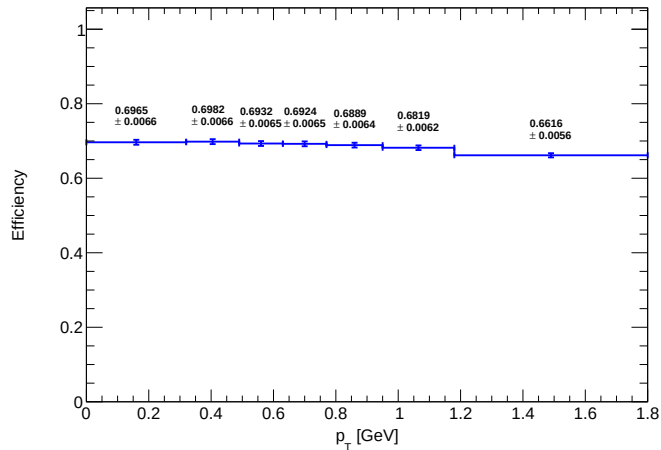
Y total reco LD2, Y mix reco LD2

Flask:

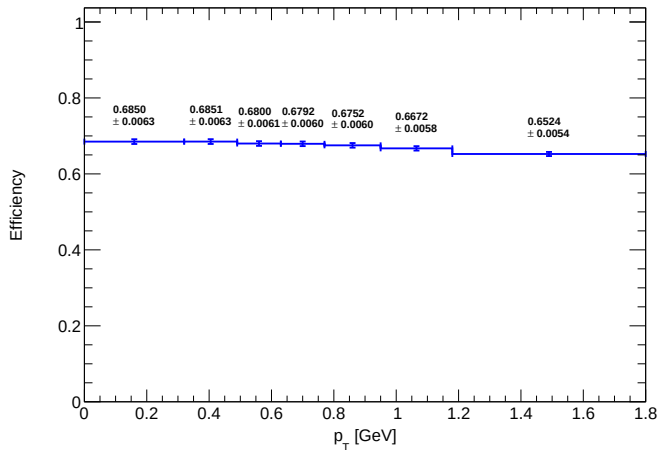
Y total reco Flask, Y mix reco Flask

Note: The step-by-step procedure was explained in docDB: 11494-v5

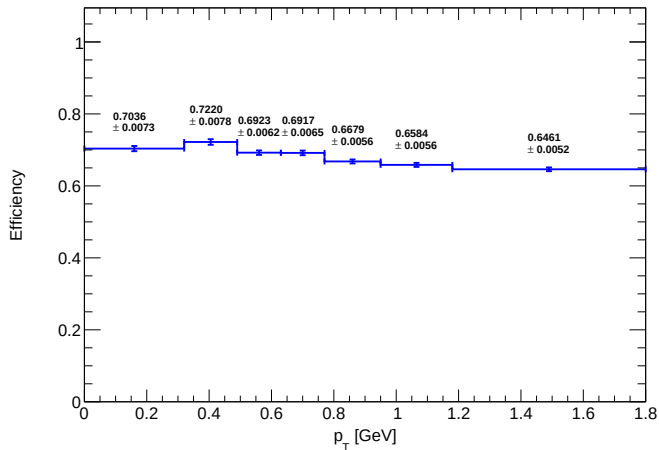
Avg Reco Eff (Total) (LH2)



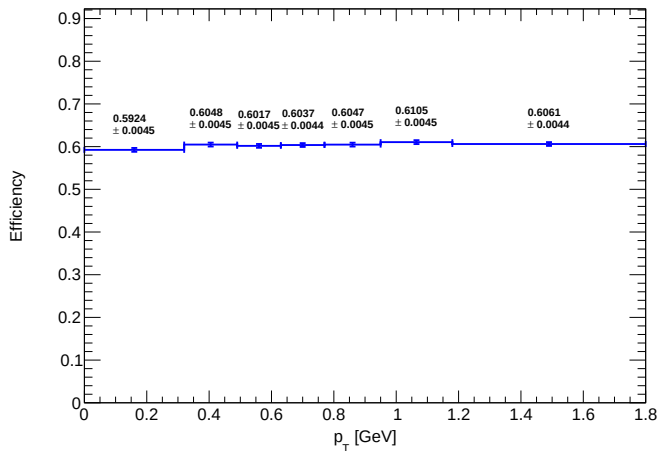
Avg Reco Eff (Total) (LD2)



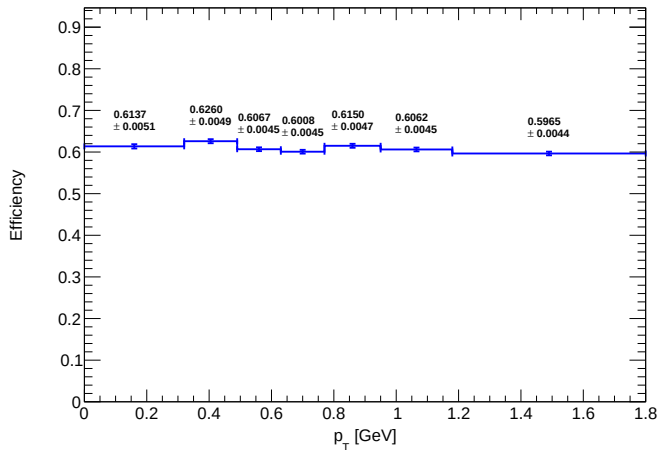
Avg Reco Eff (Total) (Flask)



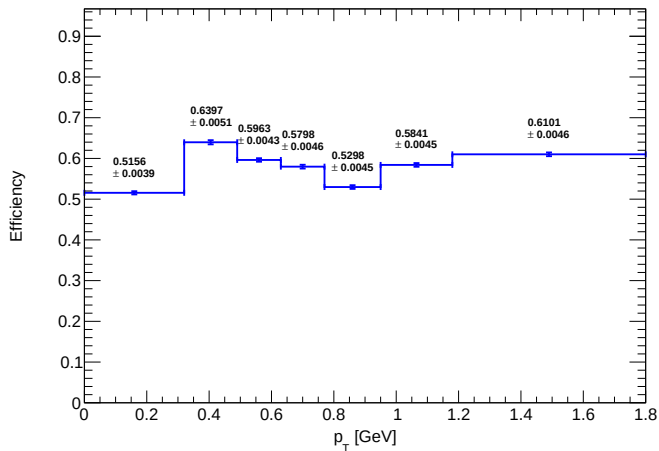
Avg Reco Eff (Mix) (LH2)



Avg Reco Eff (Mix) (LD2)



Avg Reco Eff (Mix) (Flask)



Determination of Hodoscope Efficiencies

Following hodoscope efficiencies with the propagated uncertainties were calculated for targets: [LH2, LD2 and Flask] by using the latest hodoscope paddle efficiency table:

LH2 Target:

E total hodo LH2, E mix hodo LH2

LD2 Target:

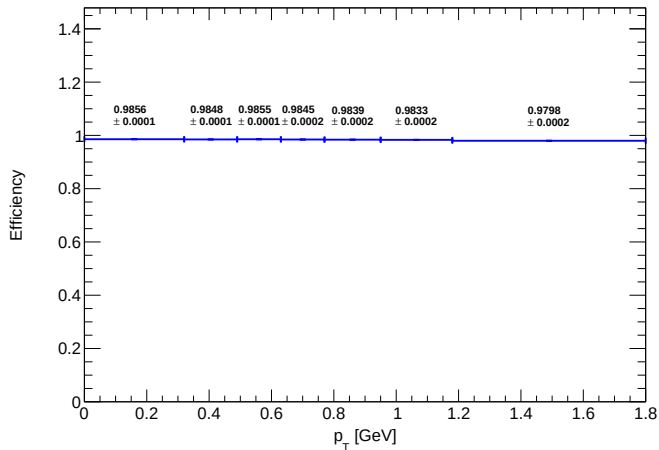
E total hodo LD2, E mix hodo LD2

Flask:

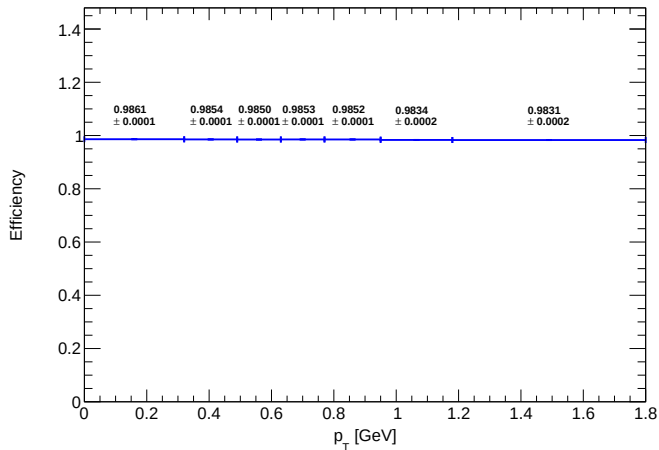
E total hodo Flask, E mix hodo Flask

Note: The latest hodoscope paddle efficiency calculations can be found in docDB: 11467-v10

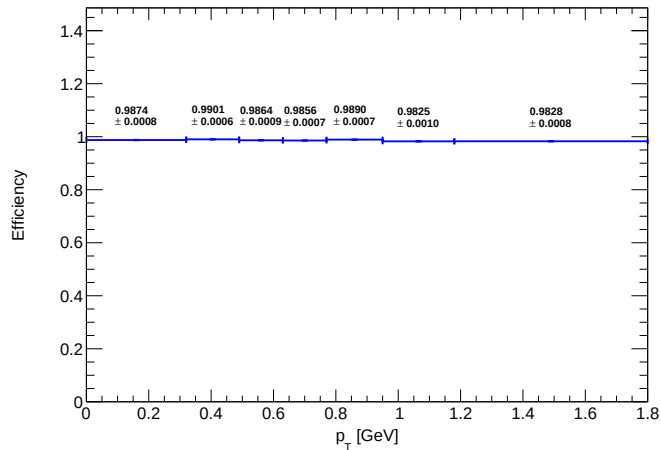
Avg Hodo Eff (Total) (LH2)



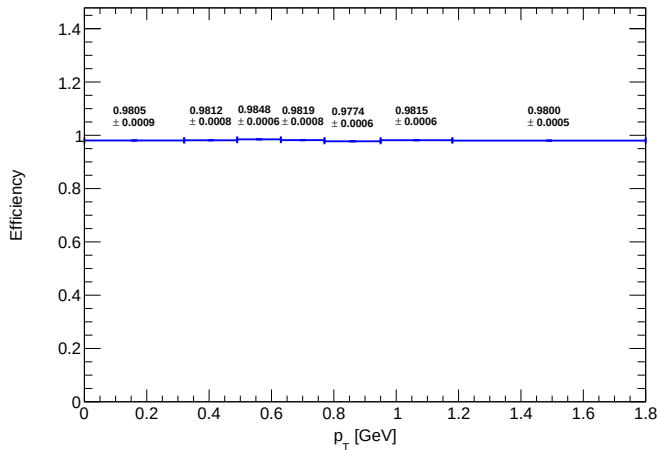
Avg Hodo Eff (Total) (LD2)



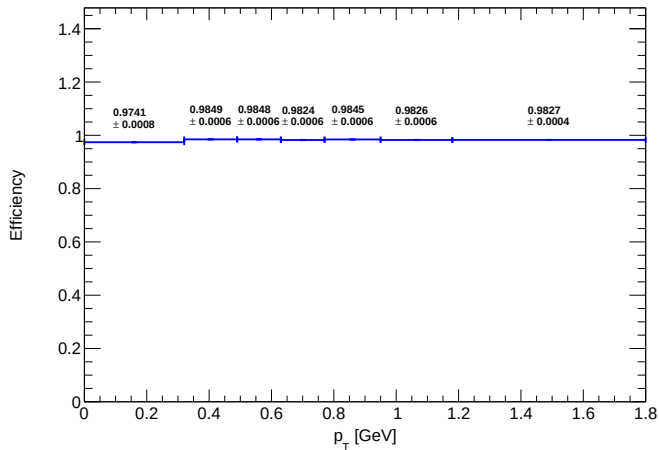
Avg Hodo Eff (Total) (Flask)



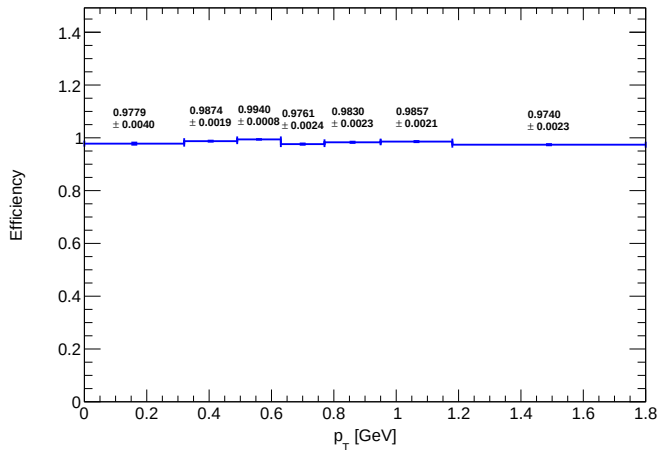
Avg Hodo Eff (Mix) (LH2)



Avg Hodo Eff (Mix) (LD2)



Avg Hodo Eff (Mix) (Flask)



Determination of final efficiencies

Now that we have calculated both average reco. and average hodo. efficiencies, by taking the product, final efficiencies can be calculated with the correct propagated uncertainties for the targets: [LH2, LD2, Flask]

LH2 Target:

E total final LH2, E mix final LH2

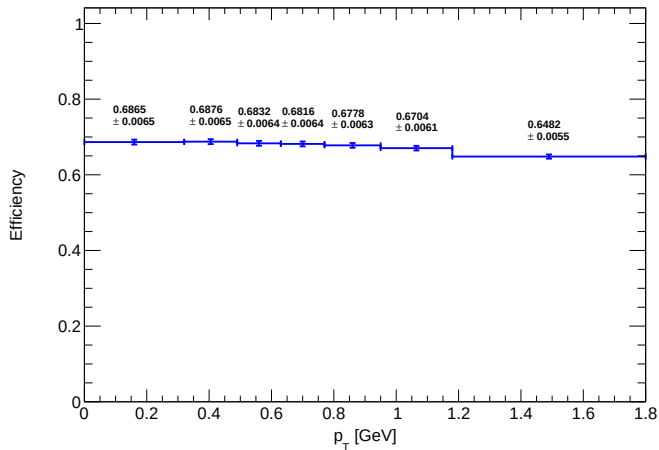
LD2 Target:

E total final LD2, E mix final LD2

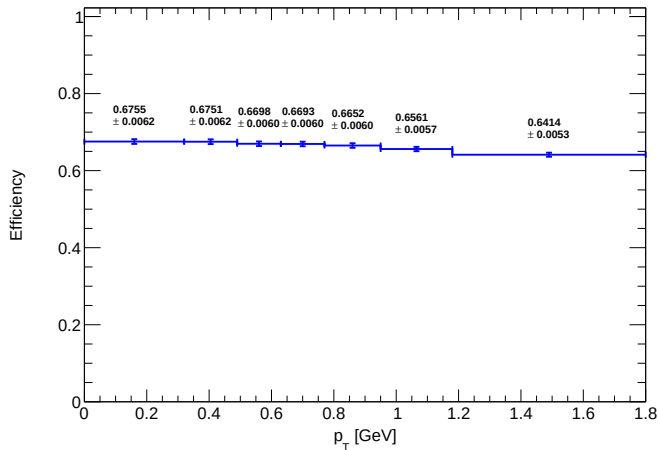
Flask:

E total final Flask, E mix final Flask

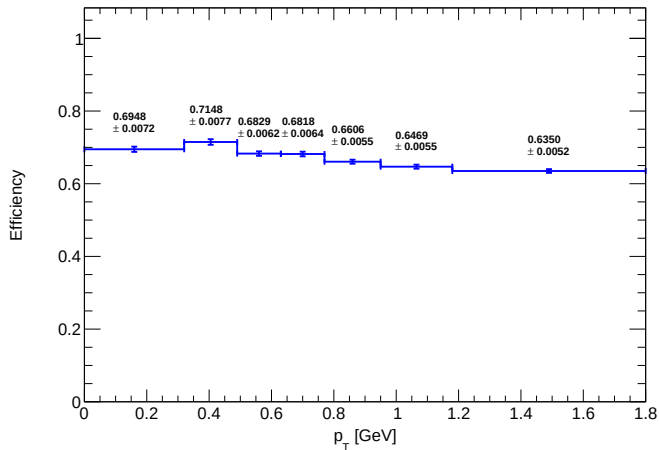
Avg Final Eff (Total) (LH2)

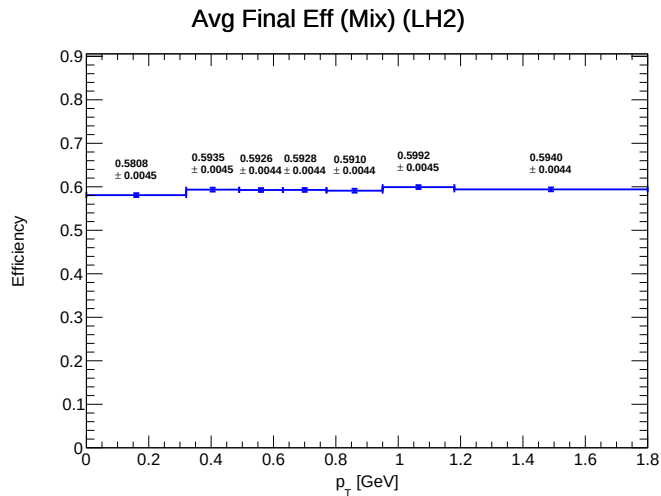


Avg Final Eff (Total) (LD2)

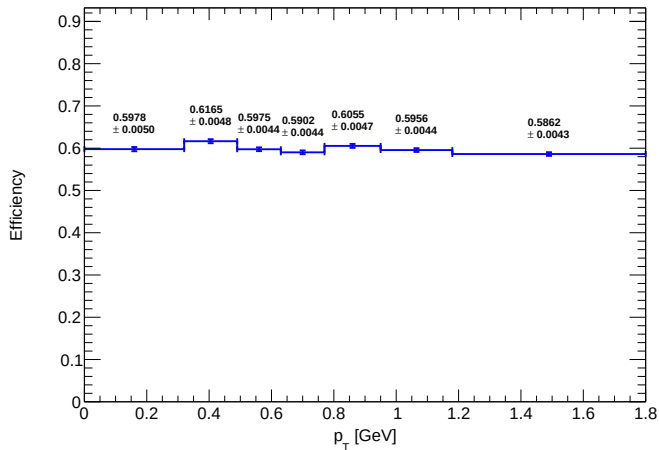


Avg Final Eff (Total) (Flask)

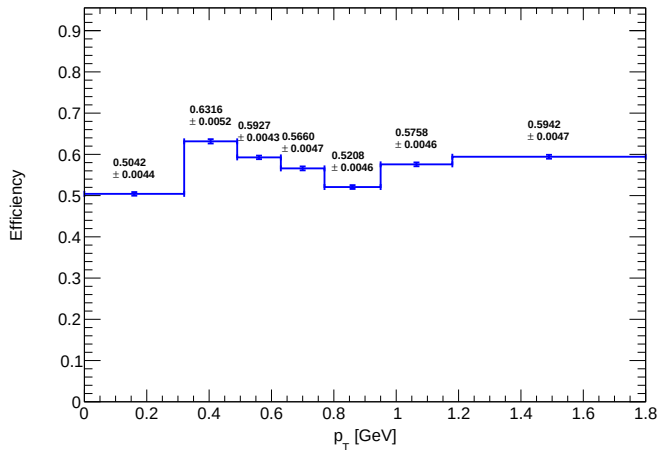




Avg Final Eff (Mix) (LD2)



Avg Final Eff (Mix) (Flask)

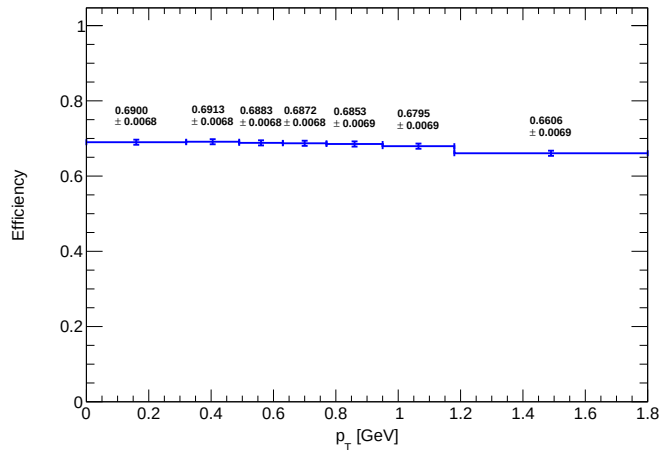


Determination of ϵ_{signal}

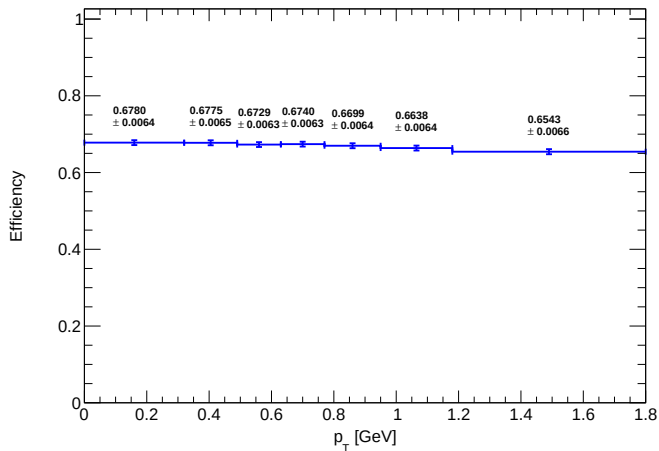
Now that we have calculated Y_{total} , Y_{mix} , $\langle\epsilon\rangle_{\text{total}}$ and $\langle\epsilon\rangle_{\text{mix}}$, it is possible to calculate ϵ_{signal} using following formula:

$$\epsilon_{\text{signal}} = \frac{\langle\epsilon\rangle_{\text{total}} Y_{\text{total}} - \langle\epsilon\rangle_{\text{mix}} Y_{\text{mix}}}{Y_{\text{total}} - Y_{\text{mix}}}$$

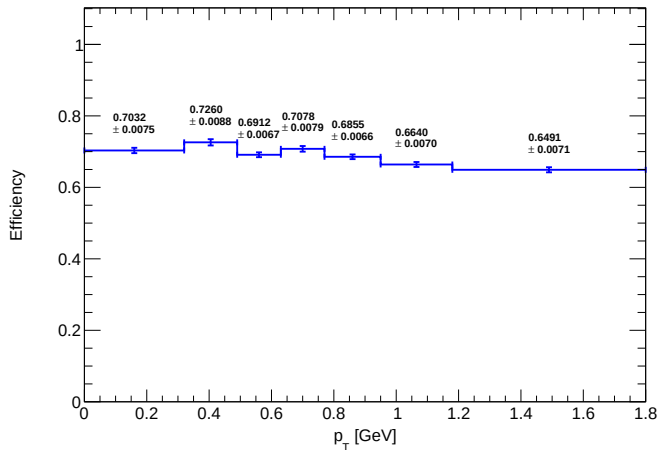
Avg Signal Efficiency (LH2)



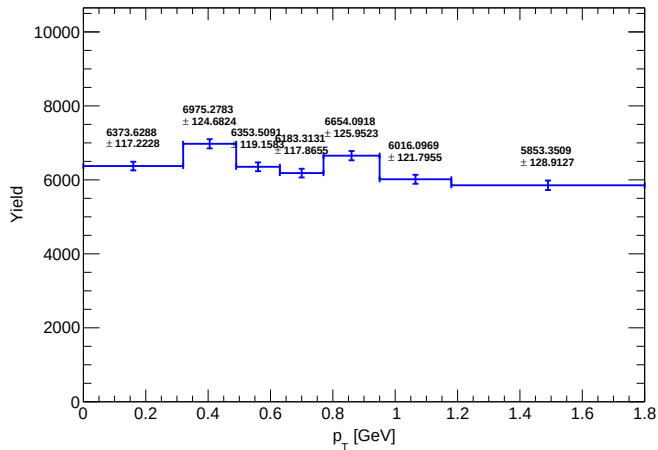
Avg Signal Efficiency (LD2)



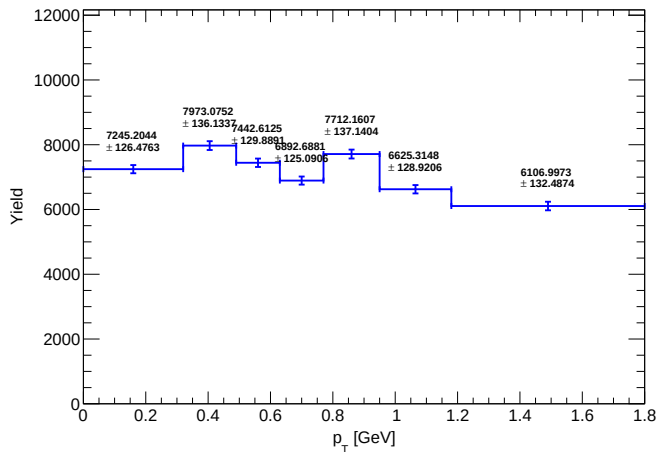
Avg Signal Efficiency (Flask)



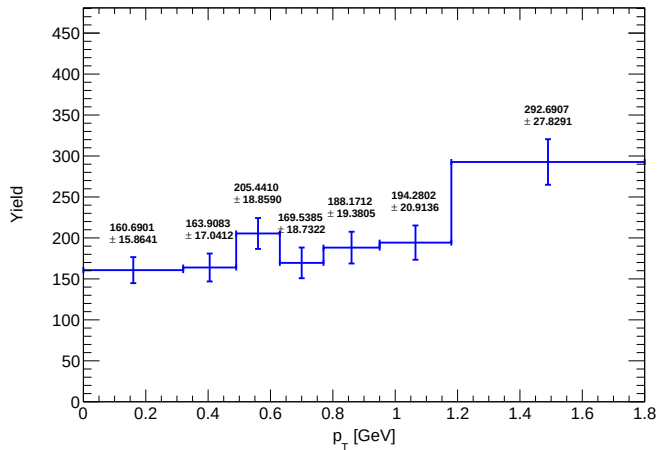
Corrected Yield (Total Error) (LH2)



Corrected Yield (Total Error) (LD2)



Corrected Yield (Total Error) (Flask)



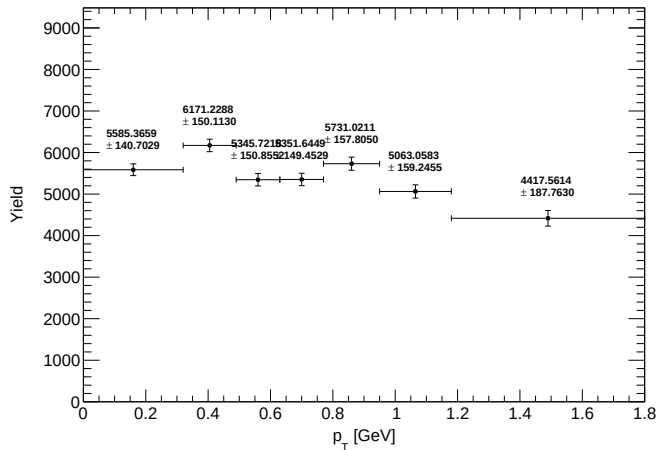
Determination of Signal candidates

Now that we have calculated Y_{total} , Y_{mix} and ϵ_{signal} it is possible to calculate signal candidates for both LH2 and LD2 targets using following equations:

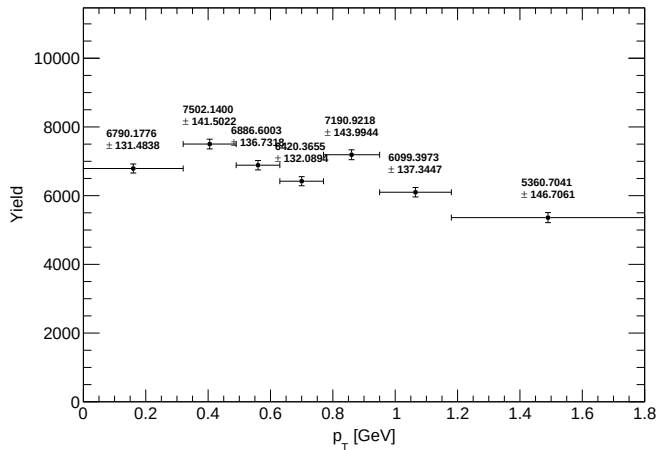
$$Y_{corrected}^{LH2} = \frac{Y_{total}^{LH2} - Y_{mixed}^{LH2}}{\langle \epsilon_{signal}^{LH2} \rangle} - \frac{I_{LH2}}{I_{flask}} \left(\frac{Y_{total}^{flask} - Y_{mixed}^{flask}}{\langle \epsilon_{signal}^{flask} \rangle} \right) \quad (1)$$

$$Y_{corrected}^{LD2} = \frac{Y_{total}^{LD2} - Y_{mixed}^{LD2}}{\langle \epsilon_{signal}^{LD2} \rangle} - \frac{I_{LD2}}{I_{flask}} \left(\frac{Y_{total}^{flask} - Y_{mixed}^{flask}}{\langle \epsilon_{signal}^{flask} \rangle} \right) - \frac{T_{HD}}{T_{HH}} \frac{I_{LD2}}{I_{LH2}} \left[\frac{Y_{total}^{LH2} - Y_{mixed}^{LH2}}{\langle \epsilon_{signal}^{LH2} \rangle} - \frac{I_{LH2}}{I_{flask}} \left(\frac{Y_{total}^{flask} - Y_{mixed}^{flask}}{\langle \epsilon_{signal}^{flask} \rangle} \right) \right] \quad (2)$$

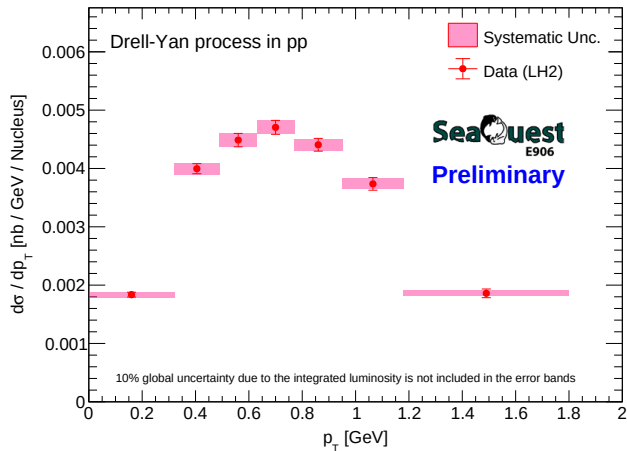
Corrected Yield (LH2 - Flask)



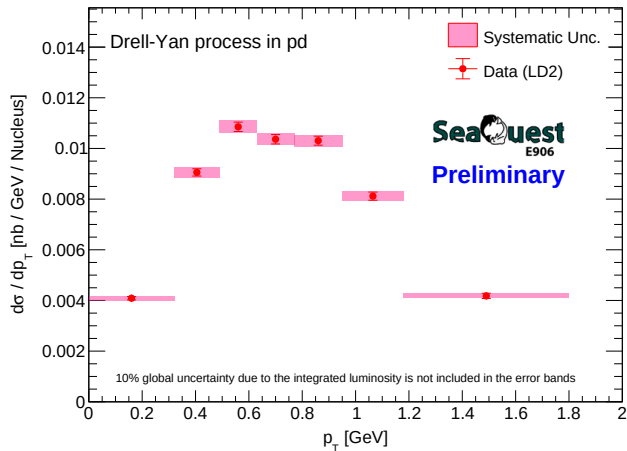
Corrected Yield (LD2 - LH2 - Flask)



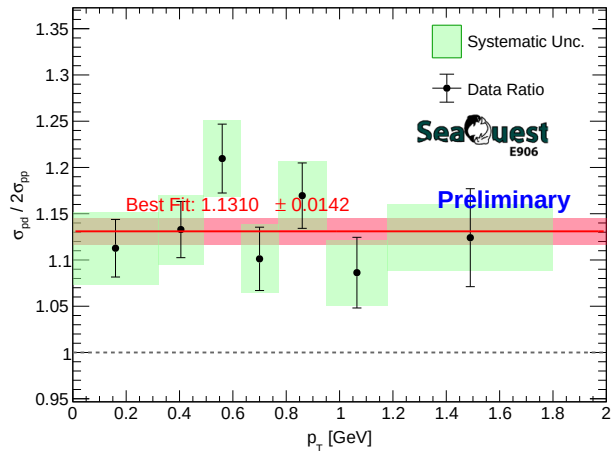
CrossSection_LH2_geom_vs_pT_with_logo



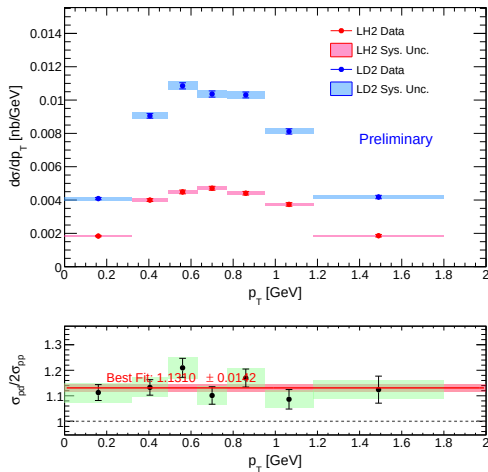
CrossSection_LD2_geom_vs_pT_with_logo



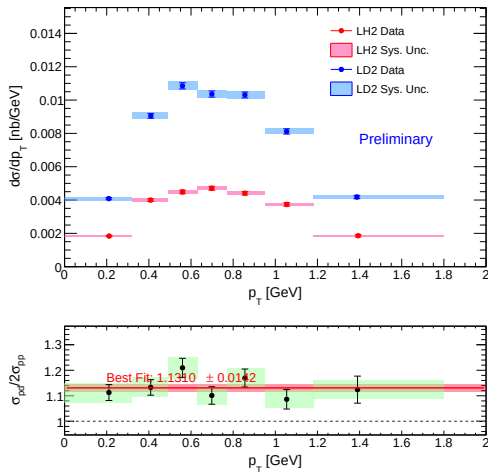
CrossSection_Ratio_pd_2pp_vs_pT_geom_with_logo



Combined_XSec_Ratio_vs_pT_geom



Combined_XSec_Ratio_vs_pT_true_pt

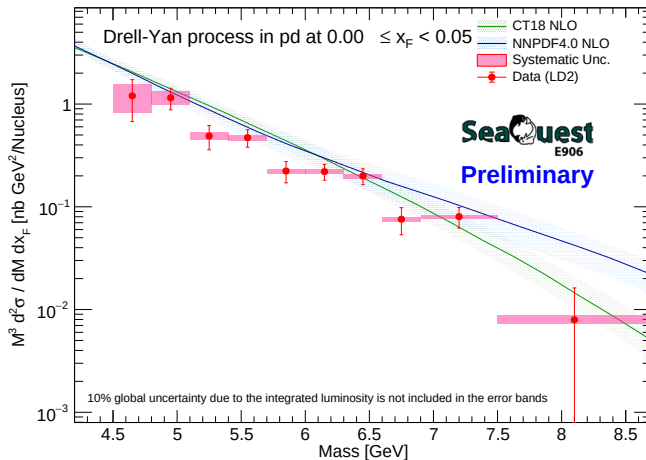


Next Steps

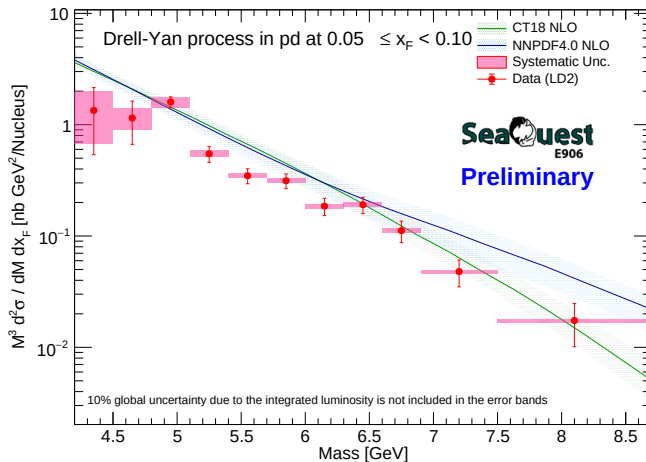
- Need to work on unfolding (work on progress).
- Planning to present preliminary plots during upcoming APS DPF meeting.
- Analysis note will be written for this Run 2-3 release, as a preliminary result.

Backup Slides

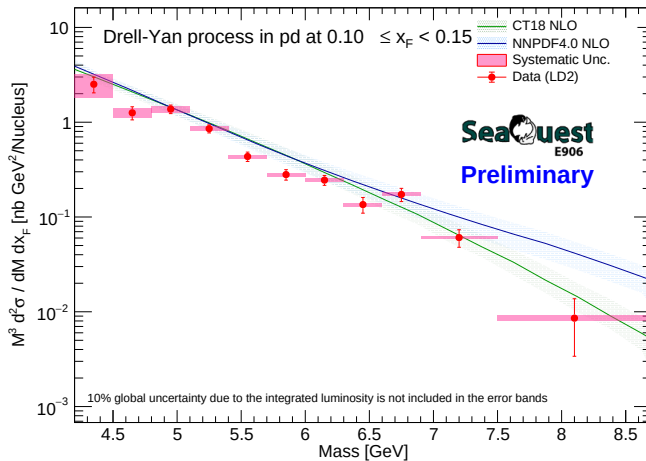
Cross-Section LD2 $0.00 \leq x_F < 0.05$



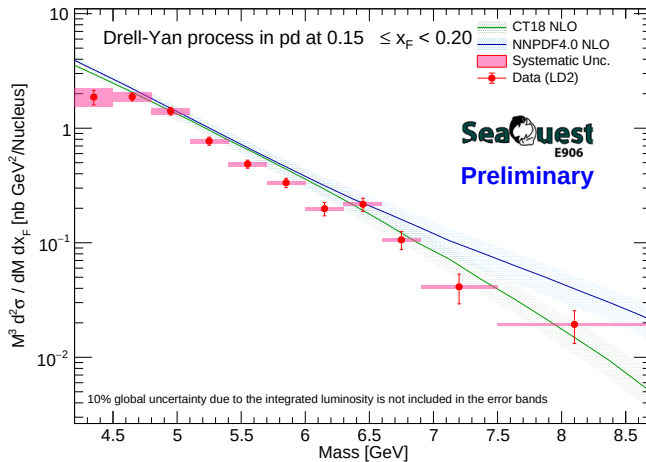
Cross-Section LD2 $0.05 \leq x_F < 0.10$



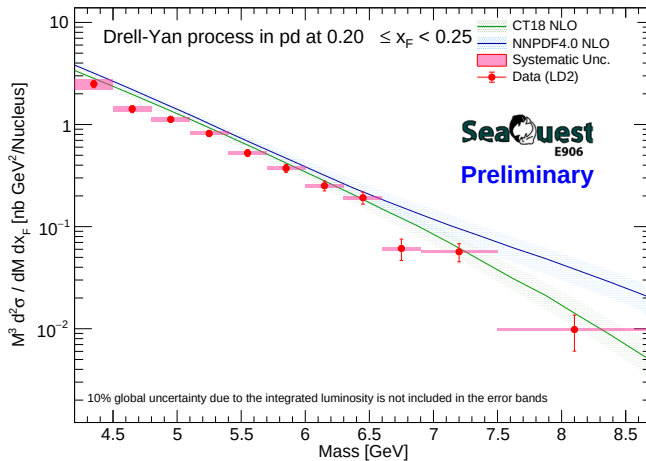
Cross-Section LD2 $0.10 \leq x_F < 0.15$



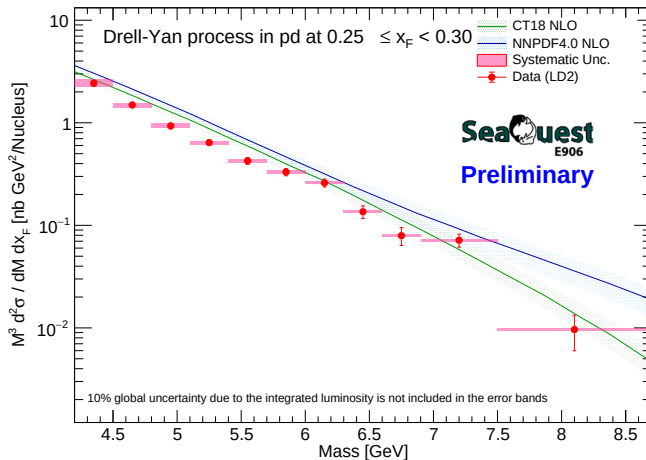
Cross-Section LD2 $0.15 \leq x_F < 0.20$



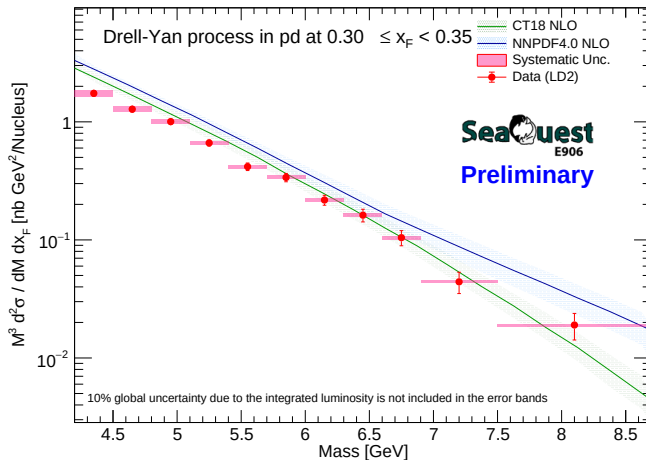
Cross-Section LD2 $0.20 \leq x_F < 0.25$



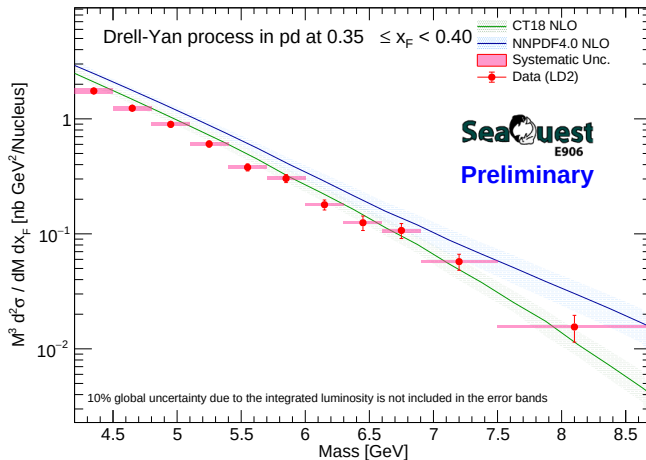
Cross-Section LD2 $0.25 \leq x_F < 0.30$



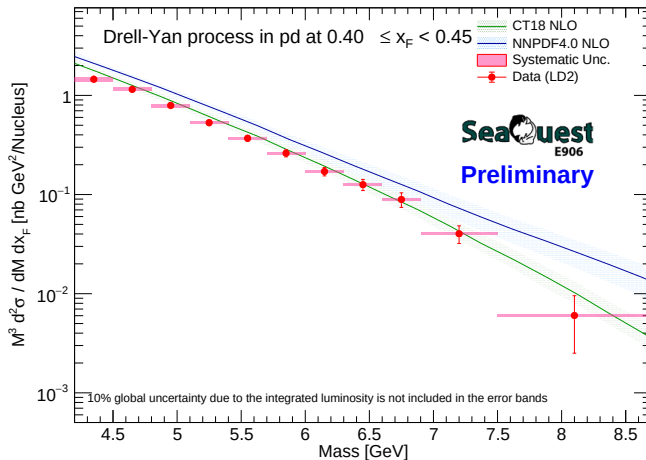
Cross-Section LD2 $0.30 \leq x_F < 0.35$



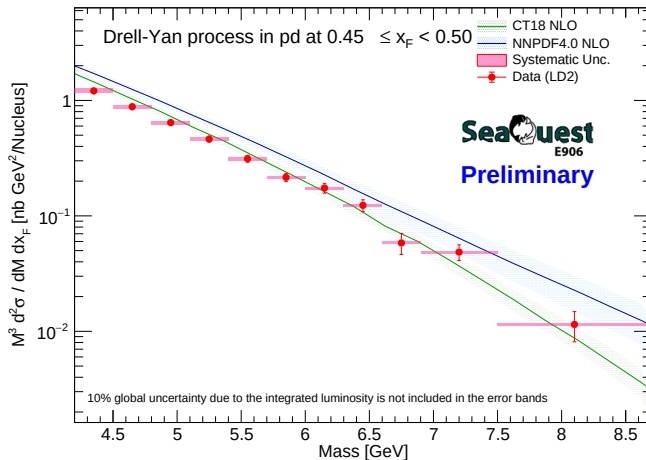
Cross-Section LD2 $0.35 \leq x_F < 0.40$



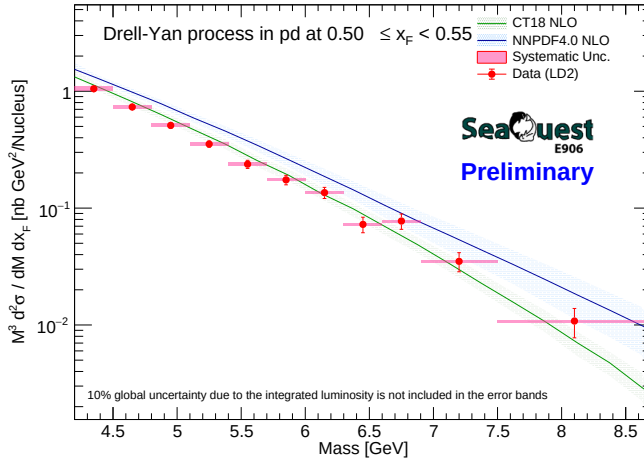
Cross-Section LD2 $0.40 \leq x_F < 0.45$



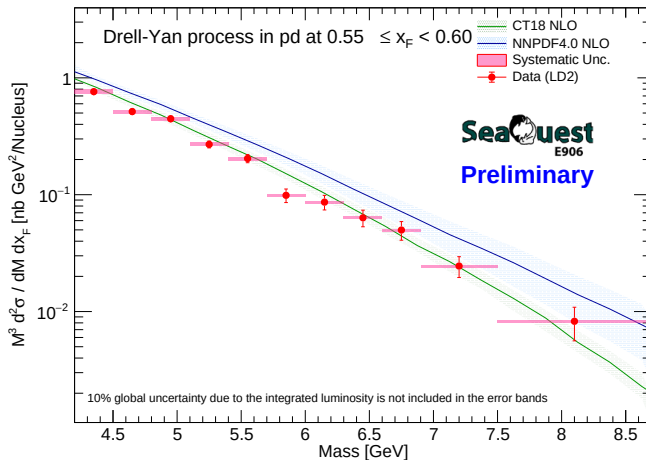
Cross-Section LD2 $0.45 \leq x_F < 0.50$



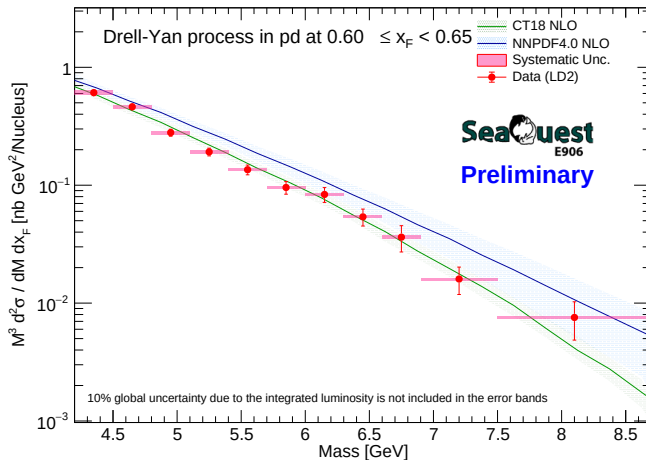
Cross-Section LD2 $0.50 \leq x_F < 0.55$



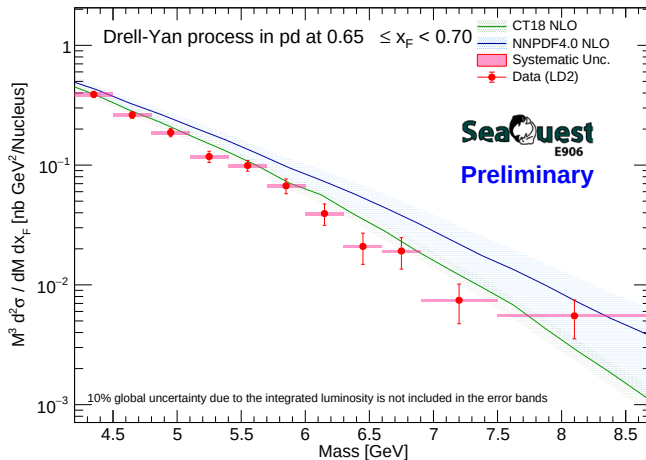
Cross-Section LD2 $0.55 \leq x_F < 0.60$



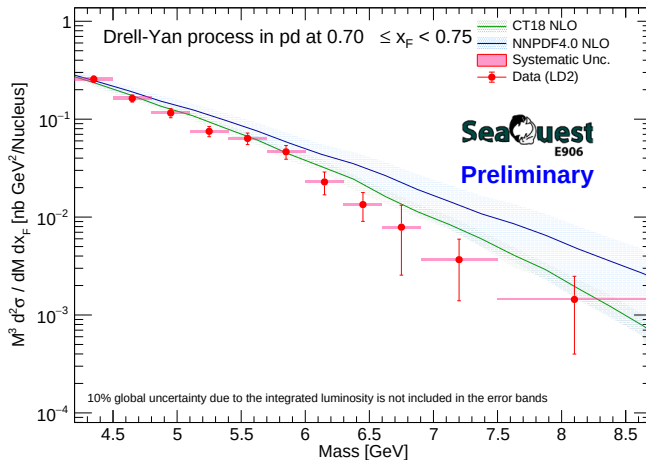
Cross-Section LD2 $0.60 \leq x_F < 0.65$



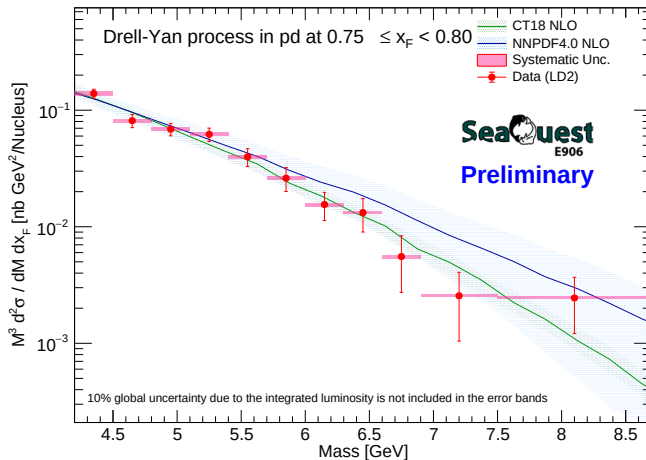
Cross-Section LD2 $0.65 \leq x_F < 0.70$



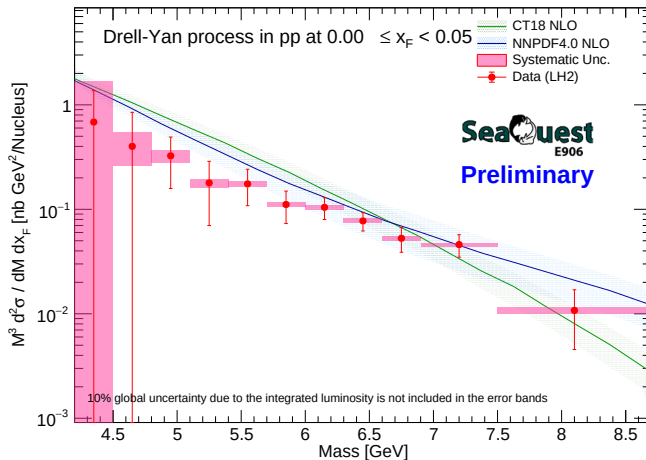
Cross-Section LD2 $0.70 \leq x_F < 0.75$



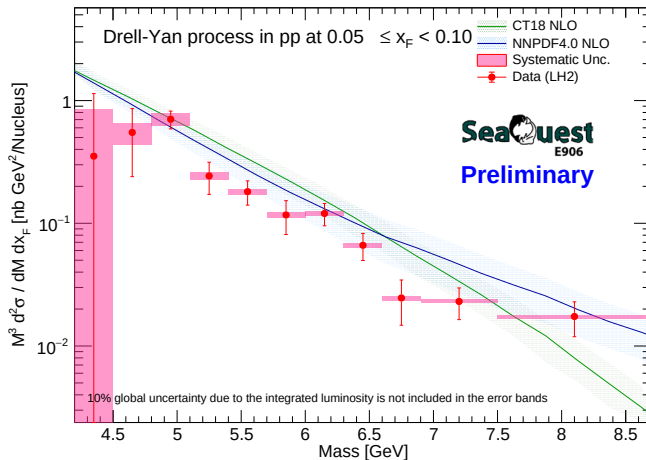
Cross-Section LD2 $0.75 \leq x_F < 0.80$



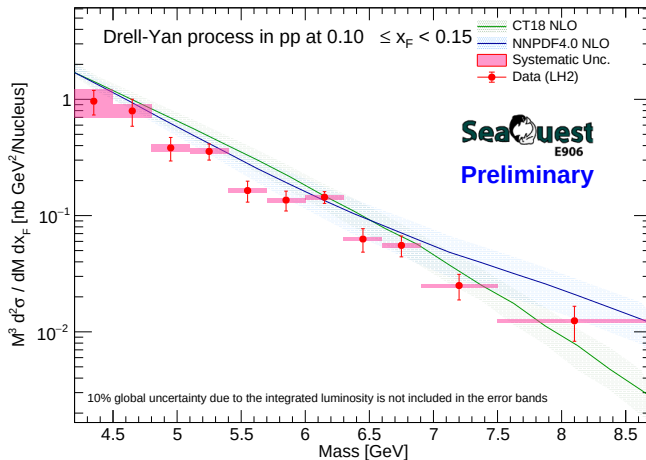
Cross-Section LH2 $0.00 \leq x_F < 0.05$



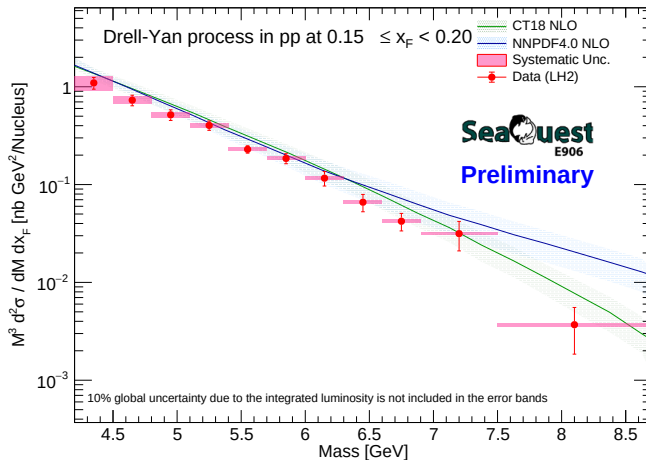
Cross-Section LH2 $0.05 \leq x_F < 0.10$



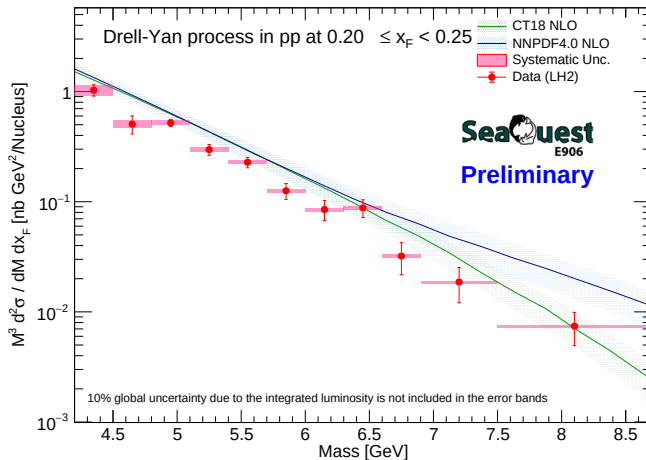
Cross-Section LH2 $0.10 \leq x_F < 0.15$



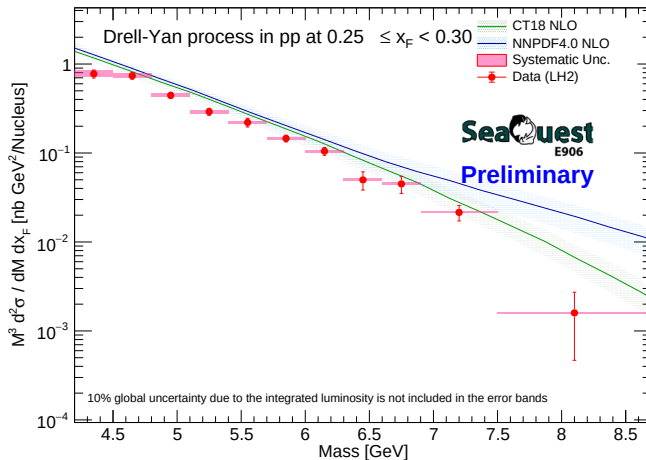
Cross-Section LH2 $0.15 \leq x_F < 0.20$



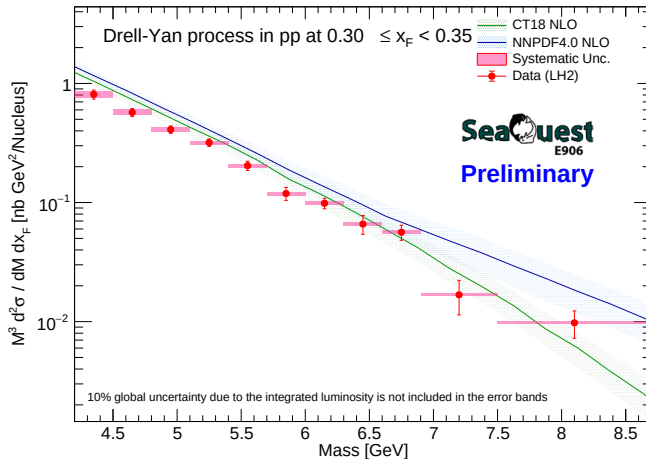
Cross-Section LH2 $0.20 \leq x_F < 0.25$



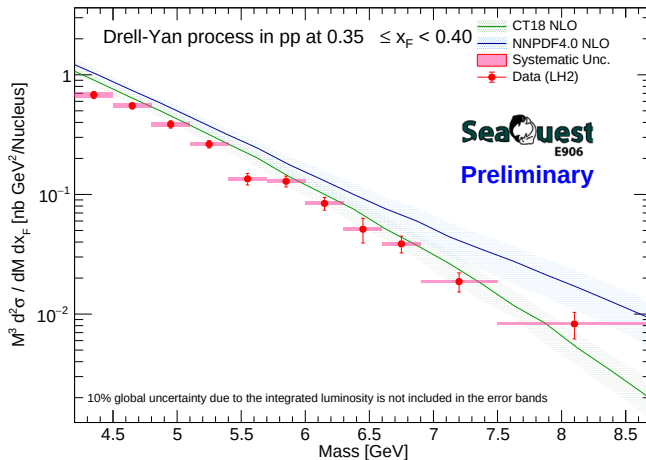
Cross-Section LH2 $0.25 \leq x_F < 0.30$



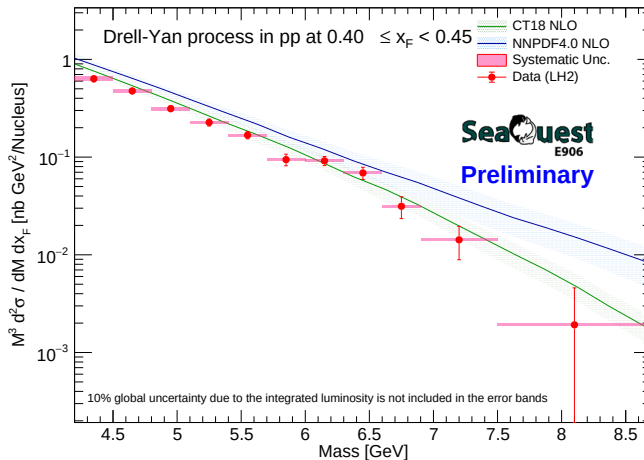
Cross-Section LH2 $0.30 \leq x_F < 0.35$



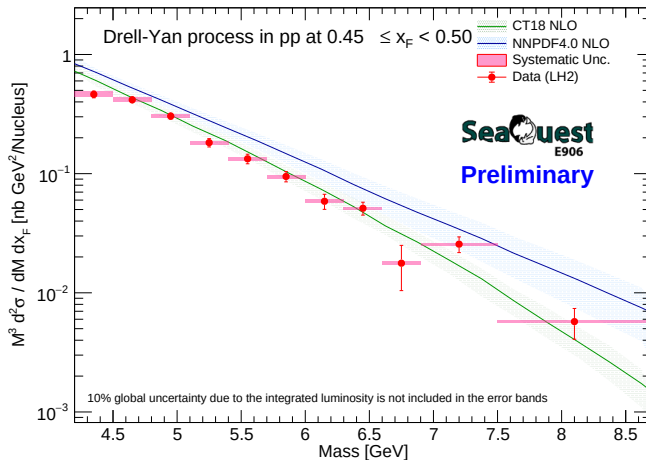
Cross-Section LH2 $0.35 \leq x_F < 0.40$



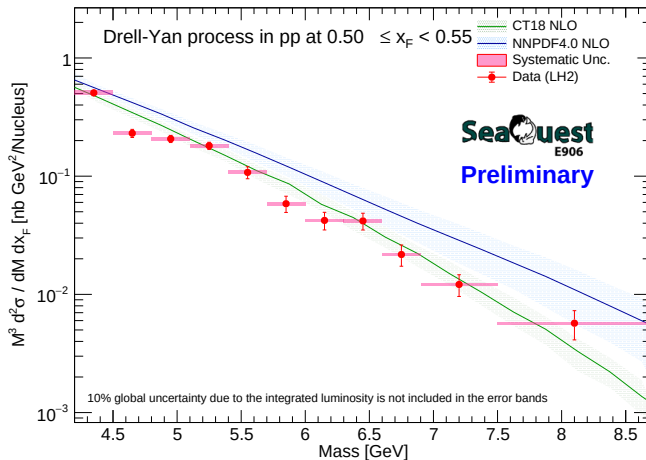
Cross-Section LH2 $0.40 \leq x_F < 0.45$



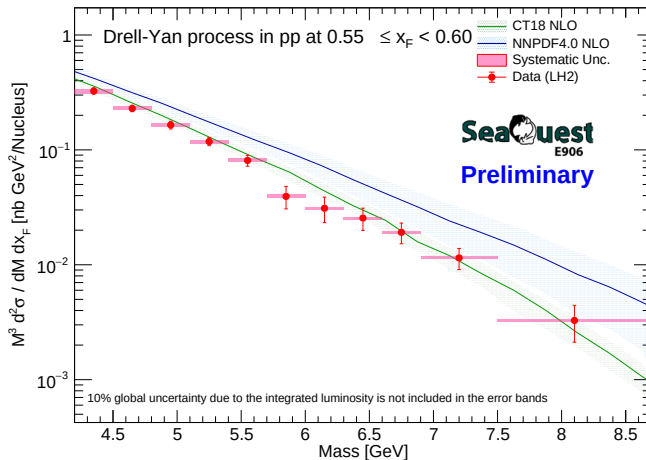
Cross-Section LH2 $0.45 \leq x_F < 0.50$



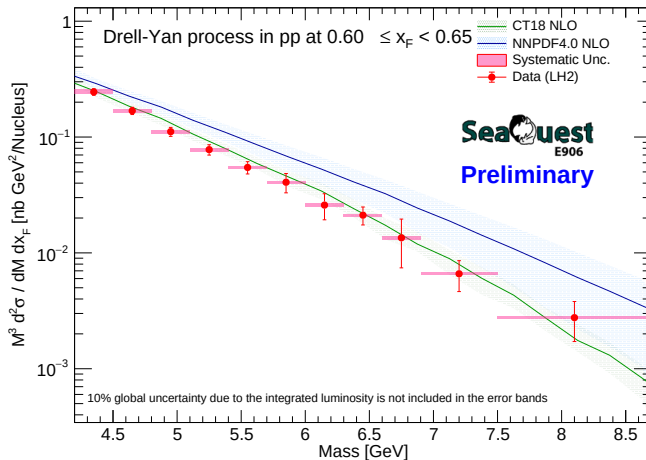
Cross-Section LH2 $0.50 \leq x_F < 0.55$



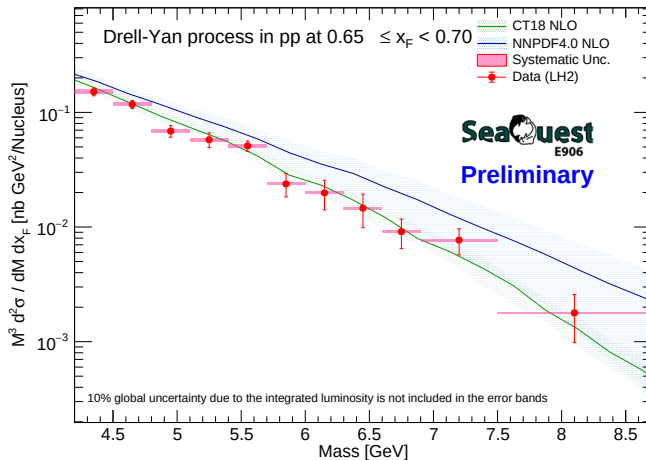
Cross-Section LH2 $0.55 \leq x_F < 0.60$



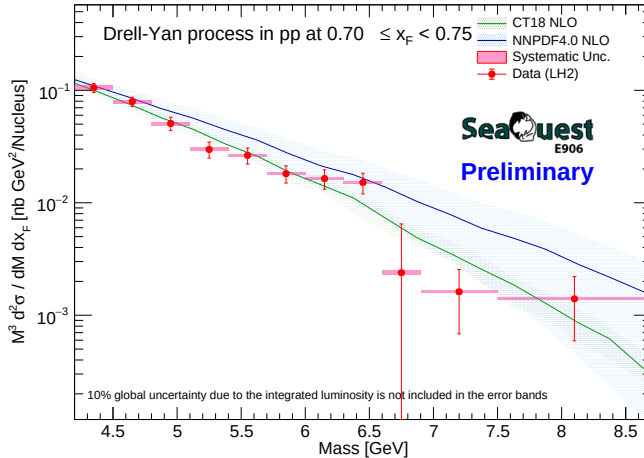
Cross-Section LH2 $0.60 \leq x_F < 0.65$



Cross-Section LH2 $0.65 \leq x_F < 0.70$



Cross-Section LH2 $0.70 \leq x_F < 0.75$



Cross-Section LH2 $0.75 \leq x_F < 0.80$

